

Analysis of a new chaotic system

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Abstract

This paper reports a new three-dimensional continuous quadratic autonomous chaotic system, modified from the Lorenz system, in which each equation contains a single quadratic cross-product term, which is different from the Lorenz, Rössler, Chen, Lü systems. Basic properties of the new system are analyzed by means of Lyapunov exponent spectrum and bifurcation diagrams. Analysis results show that this system has complex dynamics with some interesting characteristics.

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1. Introduction

In 1963, Lorenz found the first chaotic attractor in a three-dimensional (3D) autonomous system when he studied the atmospheric convection [1]. As the first

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chaotic model, the Lorenz system has become a paradigm for chaos research, which is described by

$$\begin{aligned}\dot{x} &= a(y - x), \\ \dot{y} &= cx - y - xz, \\ \dot{z} &= xy - bz.\end{aligned}\tag{1}$$

This now-classic Lorenz system has been extensively studied in the fields of chaos theory, dynamical systems as well as chaos control and synchronization. Today, one may wonder if the Lorenz system was just a lucky instance of discovery, and what else there might have been found around this kind of amazing chaotic system. Later, Rössler [2] constructed an even simpler 3D chaotic system, written as

$$\begin{aligned}\dot{x} &= -(y + x), \\ \dot{y} &= x + ay, \\ \dot{z} &= zx - cz + b,\end{aligned}\tag{2}$$

where $a = b = 0.2$, and taking c as one of following values:

$$2, 2.3, 3.5, 4.7, 5.0, 5.7, 6, 7, 8, 9, 10, 11.$$

Clearly, only the third equation contains a cross-product term in the Rössler system (2), which therefore is not comparable with the Lorenz system (1).

Notably, over the last two decades chaos in engineering systems, such as nonlinear circuits, has gradually moved from simply being a scientific curiosity to a promising subject with practical significance and applications. Chaos is found to be useful or has great potential application in many disciplines such as in liquid mixing with low power consumption, high-performance circuit design for telecommunications, collapse prevention of power systems, biomedical engineering applications to the human brain and heart [3].

Recently, it has been noticed that purposefully creating chaos can be a key issue in many technological applications such as communication, encryption, information storage, etc. In 1999, Chen constructed a 3-D chaotic system via a simple state feedback to the second equation in the Lorenz system, yielding [4,5,10]

$$\begin{aligned}\dot{x} &= a(y - x), \\ \dot{y} &= (c - a)x + cy - xz, \\ \dot{z} &= xy - bz.\end{aligned}\tag{3}$$

Like the Lorenz system, the Chen system and later the Lü system [6] contain cross-product terms in the second and third equations. Despite its seemingly similar structure with the Lorenz system, they are topologically not equivalent [5]. Mathematically, it has already been proved that Chen's attractor does exist [7]. Chen's system was proved to be the *dual system* to the Lorenz system in the sense defined by Čelikovský and Vaněček [8]: If this type of systems is written as a sum of the linear part Ax and the quadratic part $f(x)$, i.e., $\dot{x} = Ax + f(x)$ with $A = [a_{ij}]_{3 \times 3}$, then for the linear part, the Lorenz system satisfies $a_{12}a_{21} > 0$, while the Chen system satisfies $a_{12}a_{21} < 0$. In 2002, Lü and Chen [6] further found a chaotic system, named

the Lü system hereafter, which satisfies $a_{12}a_{21} = 0$ and represents the transition between the Lorenz system and the Chen system. This system is described by

$$\begin{aligned}\dot{x} &= a(y - x), \\ \dot{y} &= cy - xz, \\ \dot{z} &= xy - bz.\end{aligned}\quad (4)$$

Lorenz, Chen and Lü systems are similar but nonequivalent. Soon thereafter, in 2002, a very large and rich family of the so-called *Lorenz system family* was constructed [9] as a connection of the Lorenz, Lü and Chen systems; it is described by

$$\begin{aligned}\dot{x} &= (25\alpha + 10)(y - x), \\ \dot{y} &= (28 - 35\alpha)x + (29\alpha - 1)y - xz, \\ \dot{z} &= xy - \frac{1}{3}(\alpha + 8)z,\end{aligned}\quad (5)$$

where α is a real parameter, and for all $\alpha \in [0, 1]$ the system is chaotic.

In the finding of the new system, to be reported and analyzed below, one is to construct and determine the system parameter values such that the system can become chaotic following some basic ideas of chaotification [11,12], namely:

- (1) It is dissipative.
- (2) There exist some unstable equilibria, especially some saddle points.
- (3) There are some cross-product terms, so there are dynamical influence between different variables.
- (4) The system orbits are all bounded.

Under these guidelines, though not sufficient, a cross-product term is added to the first equation in the Lorenz system. It turns out that the new system has five equilibria, therefore is not topologically equivalent to the Lorenz system, Rössler system, Chen system and *Lorenz system family*. Detailed simulation and analysis results on the chaotic behaviors of this new system are given in the next two sections.

2. The new chaotic system and its basic properties

In retrospect, Chen found a new system (3) based on some chaotification analysis, and a linear feedback control term $u = -ax + (c + 1)y$ was added to the second equation of the Lorenz system.

We add a cross-product nonlinear term to the first equation of the Lorenz system, obtaining a new system as follows:

$$\begin{aligned}\dot{x} &= a(y - x) + yz, \\ \dot{y} &= cx - y - xz, \\ \dot{z} &= xy - bz,\end{aligned}\quad (6)$$

where notably each equation has one single cross-product term, different from the Lorenz system, Rössler system, Chen system and Lü system, and even Lorenz system family.

System (6) has the following basic properties:

- (1) *Symmetry* about the z -axis, which is invariant for the coordinate transformation $(x, y, z) \rightarrow (-x, -y, z)$.
- (2) *Dissipation*: Since

$$\nabla V = \frac{\partial \dot{x}}{\partial x} + \frac{\partial \dot{y}}{\partial y} + \frac{\partial \dot{z}}{\partial z} = -(a + b + 1), \quad (7)$$

as long as $a + b + 1 > 0$, system (6) is dissipative, and it converges to a set of measure zero exponentially:

$$\frac{dV}{dt} = e^{-(a+b+1)t}. \quad (8)$$

It means

$$V(t) = V_0 e^{-(a+b+1)t}$$

for initial cubage V_0 .

- (3) *Stability*: System (6) is globally, unanimously and asymptotically stable about the origin under the condition

$$c < \frac{a}{a+1}. \quad (9)$$

To verify this, take the Lyapunov function

$$V(x, y, z) = \frac{1}{2}(x^2 + (a+1)y^2 + az^2). \quad (10)$$

Then,

$$\begin{aligned} \dot{V} &= x\dot{x} + (a+1)y\dot{y} + az\dot{z} \\ &= -\frac{a+ac+c}{2}(x-y)^2 + \frac{-a+ac+c}{2}x^2 + \frac{-a+ac+c-2}{2}y^2 - abz^2. \end{aligned}$$

Hence, $\dot{V} < 0$ if

$$\begin{aligned} -a+ac+c &< 0, \\ -a+ac+c-2 &< 0. \end{aligned}$$

This means $-a+ac+c < 0$, which is condition (9).

- (4) *Equilibria*: The equilibria of system (6) can be found by solving the following system of equations:

$$a(y-x) + yz = 0, \quad cx - y - xz = 0, \quad xy - bz = 0. \quad (11)$$

Obviously, $S_1 = [0, 0, 0]$ is an equilibrium. Set

$$\begin{aligned} d &= -2ab + abc + bc^2 \\ e &= \sqrt{b^2 c^2 (-4a + a^2 + 2ac + c^2)} \\ f &= abc + bc^2 \\ x_1 &= \frac{\sqrt{2}}{2} \sqrt{\frac{d+e}{a}}, \quad x_2 = \frac{\sqrt{2}}{2} \sqrt{\frac{d-e}{a}}, \quad y_1 = \sqrt{2}bc \sqrt{\frac{d+e}{a}} \frac{a}{f+e}, \\ y_2 &= \sqrt{2}bc \sqrt{\frac{d-e}{a}} \frac{a}{f-e}, \quad z_1 = \frac{(d+e)c}{f+e}, \quad z_2 = \frac{(d-e)c}{f-e}. \end{aligned} \quad (12)$$

Other nonzero equilibria are given by

$$\begin{aligned} S_2 &= (x_1, y_1, z_1), \quad S_3 = (-x_1, -y_1, z_1), \quad S_4 = (x_2, y_2, z_2), \\ S_5 &= (-x_2, -y_2, z_2). \end{aligned} \quad (13)$$

This system has five equilibria all together. When $d - e < 0$, x_2 and y_2 are complex number, so only S_1, S_2, S_3 are real equilibria.

- (5) *Stability of the equilibria:* First, by linearizing system (6) at S_1 , one obtains three characteristic values from its Jacobian matrix at S_1 , as follows:

$$\lambda_1 = -b, \quad \lambda_{23} = -\frac{a+1}{2} \pm \frac{1}{2} \sqrt{(a+1)^2 + 4a(c-1)}. \quad (14)$$

These are identical to the Lorenz system. As long as $c > 1$, one has $\lambda_1 = -b, \lambda_2 > 0, \lambda_3 < 0$. Therefore, the zero equilibrium S_1 is an unstable saddle point of system (6).

Then, for nonzero equilibria, their stabilities can be estimated via numerical computation, which is not further discussed here.

3. Observation of chaotic and complex dynamics

3.1. Fix $a = 35$, $c = 25$, and vary b

Fig. 1(a) shows the spectrum of Lyapunov exponents of the Lorenz system (1) with respect to parameter b . Obviously, the maximum Lyapunov exponent is negative except over the initial transient time, implying that the Lorenz system (1) is attracted into a sink, as shown in Fig. 1(b). On the contrary, Figs. 2(a) and (b) show the following: When $b \in (0, 2.25]$, the maximum Lyapunov exponent almost always equals zero, implying that the new system (6) has a periodic orbit. Fig. 3(a) shows the periodic orbit when $b = 1.5$, where symbol ‘o’, ‘*’, and ‘+’ denote equilibria S_1, S_2 and S_3 , respectively. By calculating, there are three real equilibria, $S_1 = [0, 0, 0]$, $S_2 = [7.8838, 4.6445, 24.4109]$, $S_3 = [-7.8838, -4.6445, 24.4109]$, and two complex equilibria. When $b \in (2.25, 11.5]$ the system is chaotic with a positive Lyapunov exponent (for example, with $b = 7$, the phase portrait is shown in

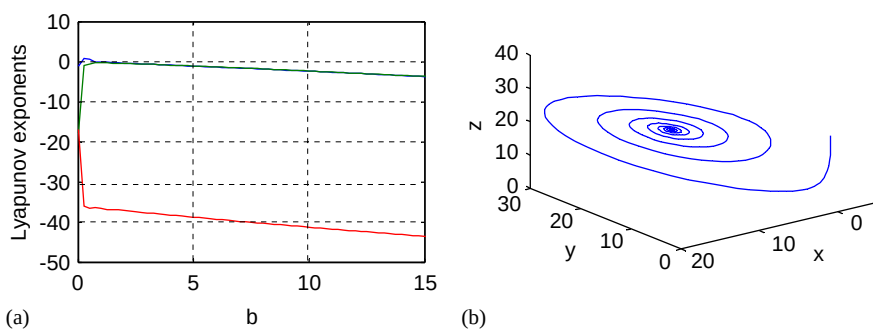


Fig. 1. Spectrum of Lyapunov exponents and phase portrait of the Lorenz system (1), $a = 35$, $c = 25$. (a) Spectrum of Lyapunov exponents w.r.t. b and (b) 3D view on the $x - y - z$ space, $b = 7$.

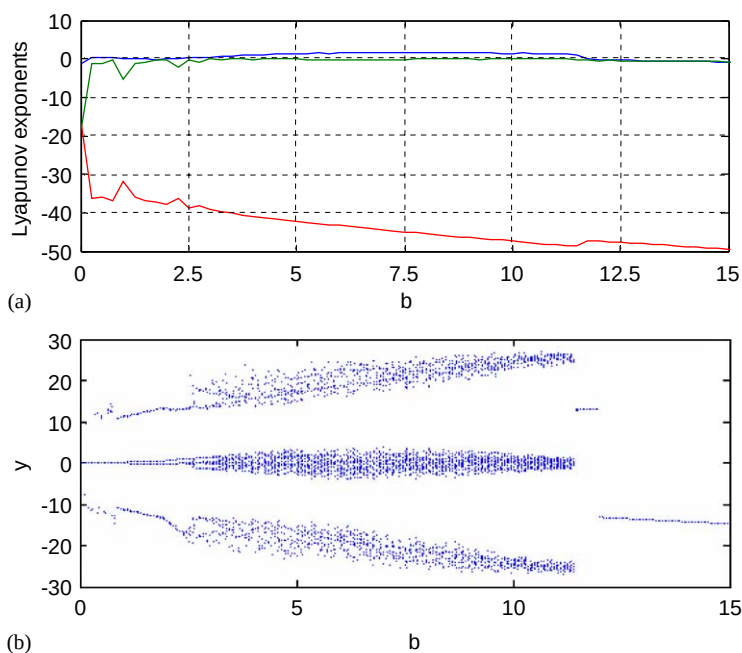


Fig. 2. Spectrum of Lyapunov exponents and bifurcation diagram of system (6) w.r.t. b , $a = 35$, $c = 25$. (a) Spectrum of Lyapunov exponents, $b \in [0, 15]$ and (b) bifurcation diagram of y , $b \in [0, 15]$.

Fig. 3(b)). In fact, there is no clear boundary between the periodic solution and chaos. From the bifurcation diagram of x_2 , shown in Fig. 2(b), one can clearly see that system (6) smoothly evolves into a chaotic region through a typical period-doubling route from the periodic orbit. When $b > 11.5$, the maximum Lyapunov exponent rapidly becomes negative and the bifurcation of x_2 abruptly changes into a line, both imply the orbit of system (6) converges to an equilibrium, shown in Fig. 3(c).

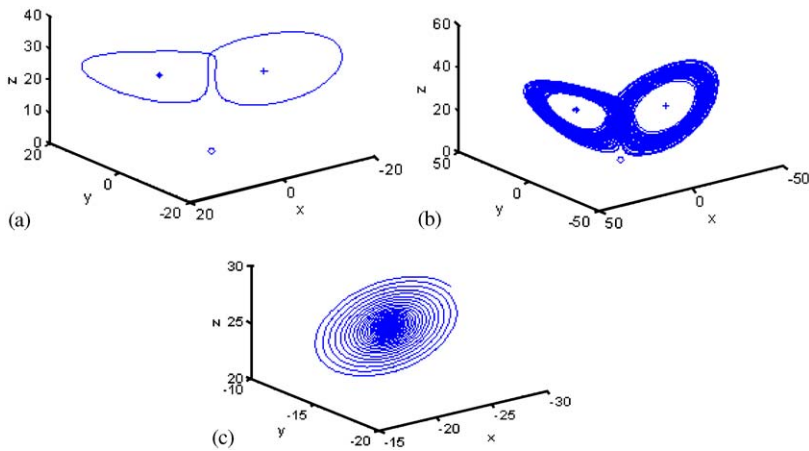


Fig. 3. Phase portraits of system (6), $a = 35$, $c = 25$. (a) 3D view on the $x - y - z$ space, $b = 1.5$, (b) 3D view on the $x - y - z$ space, $b = 7$ and (c) 3D view on the $x - y - z$ space, $b = 13$.

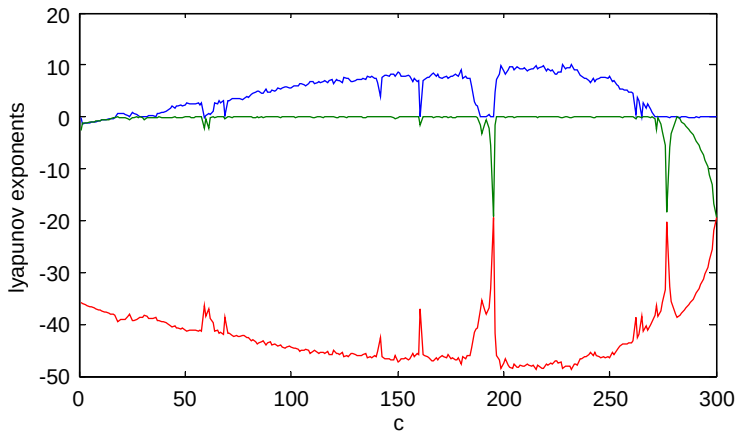


Fig. 4. Spectrum of Lyapunov exponents of system (6) w.r.t c , $a = 35$, $b = 8/3$.

The dynamics of system (6) is different from that of the Lorenz system via observing their spectra of Lyapunov exponents and bifurcation, since system (6) contains more nonlinear terms and equilibria, and brings to different locations of nonzero equilibria and different dynamical properties, even taking the same parameters.

3.2. Fix $a = 35$, $b = 8/3$, and vary c

Fig. 4 shows the spectrum of Lyapunov exponents of system (6) with respect to parameter c . Obviously, when $c \in [0, 17]$, the system is stable and converges to one

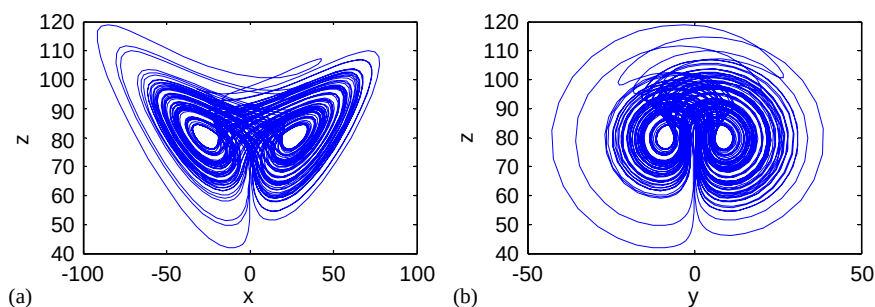


Fig. 5. Chaotic attractor of system (6), $a = 35$, $b = 8/3$, $c = 80$. (a) Projection on $x - z$ plane and (b) projection on $y - z$ plane.

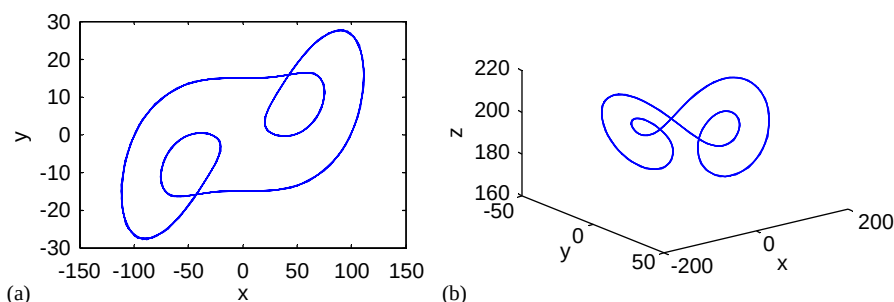
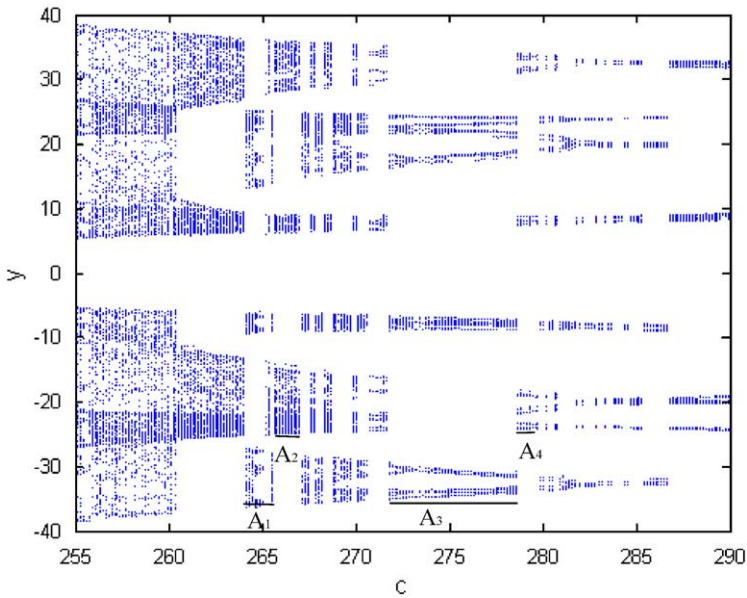
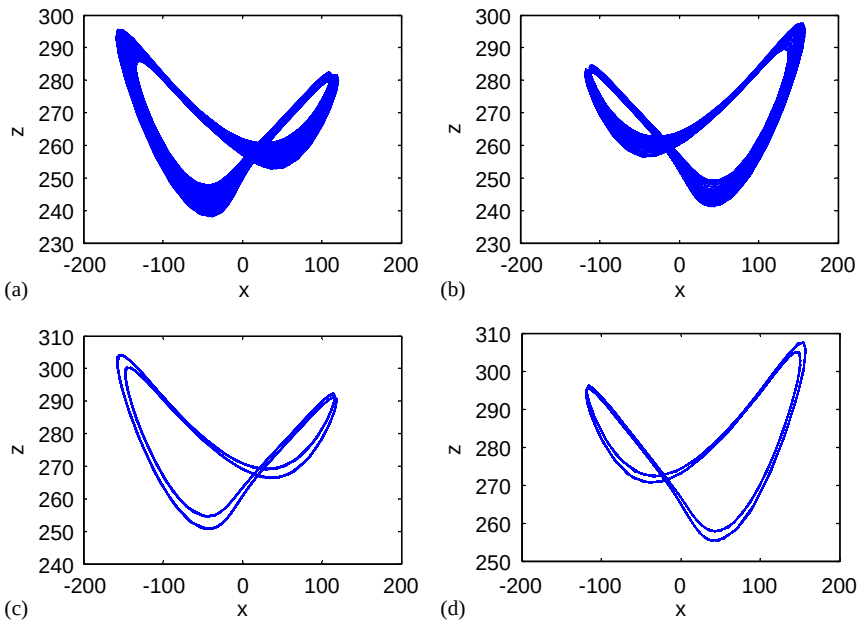


Fig. 6. Periodic orbit of system (6), $a = 35$, $b = 8/3$, $c = 193$. (a) Projection on $x - y$ plane and (b) 3D view on the $x - y - z$ space.

equilibrium. When $c \in (17, 189]$, the system is chaotic, as shown in Figs. 5(a) and (b) where $c = 80$. When $c \in (189, 194]$, the maximum Lyapunov exponent almost equals zero, which means that system (6) has a periodic orbit, as shown in Fig. 6, where $c = 193$. When $c \in (194, 271]$, the system becomes chaotic again, and during this period of time, the shape of its chaotic attractor is quite similar to the previous one when $c \in (189, 194]$. When $c > 271$, the system eventually evolves to a periodic orbit. The whole evolution process can be clearly seen from the bifurcation diagram shown in Fig. 7. The state y shows an interleaving band-like shape, which becomes quite narrow eventually as c increases, implying that the orbit of the system is eventually degenerating from chaotic to periodic. Starting with the chaotic region, with c increasing, the orbit goes through a process of reverse chaos (for example, the phase portraits are shown in Figs. 8(a) and (b)), reverse period-doubling bifurcations (Figs. 8(c) and (d)), and finally to a periodic orbit. Furthermore, notably, the position of the band-like shape of y is suddenly changed by turning into the degenerating process, as shown in divided regions A_1, A_2, A_3, A_4 , implying symmetrically similar phase portraits in shape and location, as shown in Figs. 8(a) and (b) for reverse chaos, and in Figs. 8(c) and (d) for reverse period-doubling bifurcations. In fact, the whole

Fig. 7. Bifurcation diagram of y w.r.t c , $c \in [255, 290]$.Fig. 8. Phase portraits of system (6), $a = 35$, $b = 8/3$. (a) Chaotic attractor, $c = 264.5 \in A_1$, (b) chaotic attractor, $c = 267 \in A_2$, (c) period-doubling bifurcations, $c = 275 \in A_3$ and (d) period-doubling bifurcations, $c = 279 \in A_4$.

evolution process contains two distinct processes, one side chaotic attractor is gradually degenerating to periodic, another side the degeneration process is often suddenly changed by turning and location of the system orbit.

3.3. Fix $b = 8/3$, $c = 25$, and vary a

The spectrum of Lyapunov exponents shown in Fig. 9 (to clearly see positive Lyapunov exponents, vertical axis is limited in $[-5, 5]$, so we cannot see the third

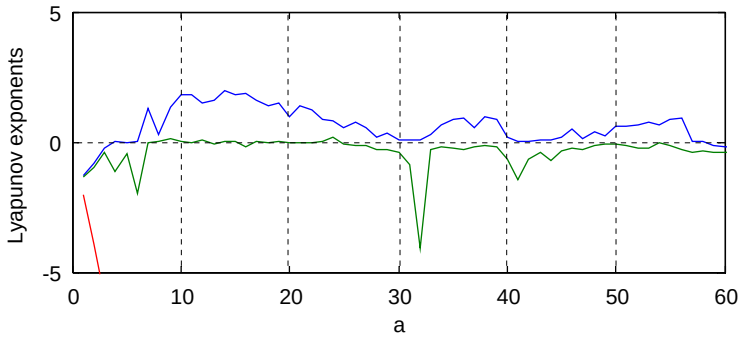


Fig. 9. Spectrum of Lyapunov exponents of system (6), $b = 8/3$, $c = 25$, $a \in [0, 60]$.

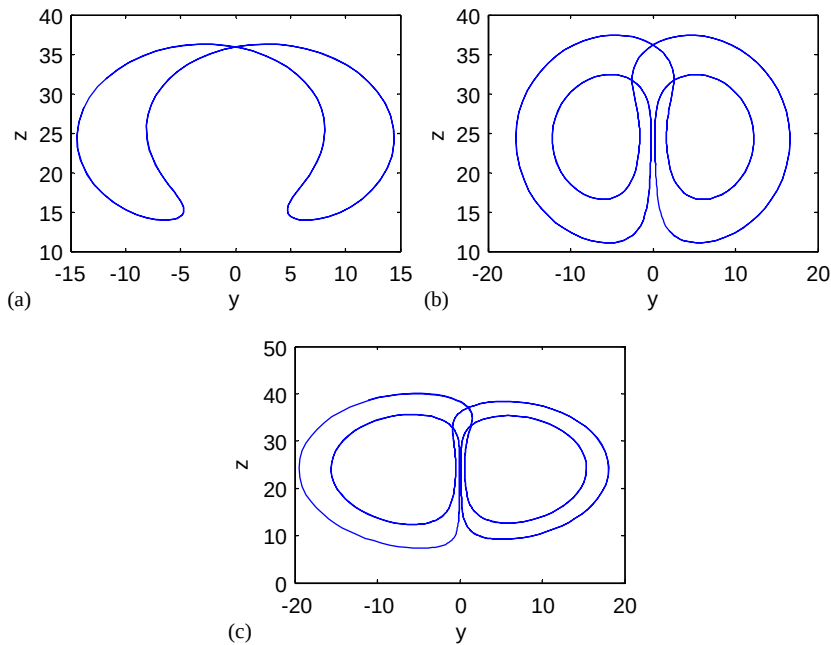


Fig. 10. Different periodic orbits of system (6), $b = 8/3$, $c = 25$. (a) Periodic solution, $a = 5$, (b) periodic solution, $a = 30$ and (c) periodic solution, $a = 45$.

curve) reveals that equilibrium point, periodic loop, and chaos appear alternately as a increases from 0 to 60. But the periodic regions do not share the same dynamic characteristics. This can be seen from Figs. 10(a)–(c), in which there are three periodic regions, but each of them has quite different periodic orbits.

Similar phenomenon can be observed from the other three chaotic regions (figures are omitted here).

3.4. Fix $b = 8/3$ and draw the division diagram on $a - c$ plane

In order to grasp the dynamic modes of system (6) with respect to parameters $a - c$, the division diagram is drawn. Varying parameters a and c within the region of $a \in [40, 70]$, $c \in [0, 130]$ by calculating Lyapunov exponents, one can observe at least four different kinds of dynamic modes in system (6). Fig. 11 displays the division of this region by a pseudo-colored map, in which the deep blue, light blue, light green, and red regions represent fixed point (F), periodic loop (P), transition (T), and chaos (C), respectively. It is clear from the figure that the dynamic mode of the system evolves from one to another as parameters a and c vary. In order to clearly show the evolving process of chaos, fix parameter $c = 104$, take $a \in [42, 60]$, and plot a line d-e-f-g-h-i, shown in Fig. 11, where $d = 42$, $e = 45.3$, $f = 46.5$, $g = 56.5$, $h = 58$, $i = 60$. Correspondingly, we draw the bifurcation diagram of state y via varying parameter along line d-e-f-g-h-i, shown in Fig. 12. The evolution process is very clearly

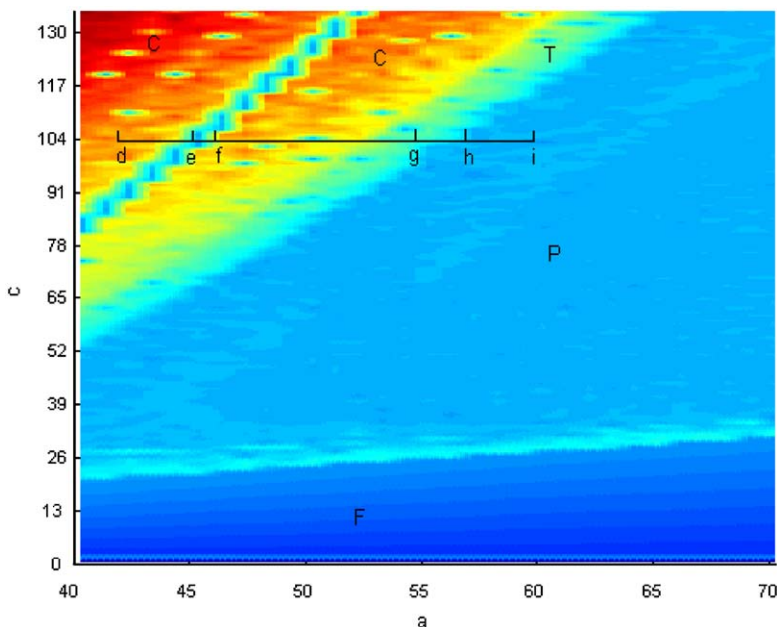


Fig. 11. Division of parameters a and c : F—equilibrium; P—periodic orbit; T—transition; C—chaos. Fix $b = 8/3$, and vary $a - c$.

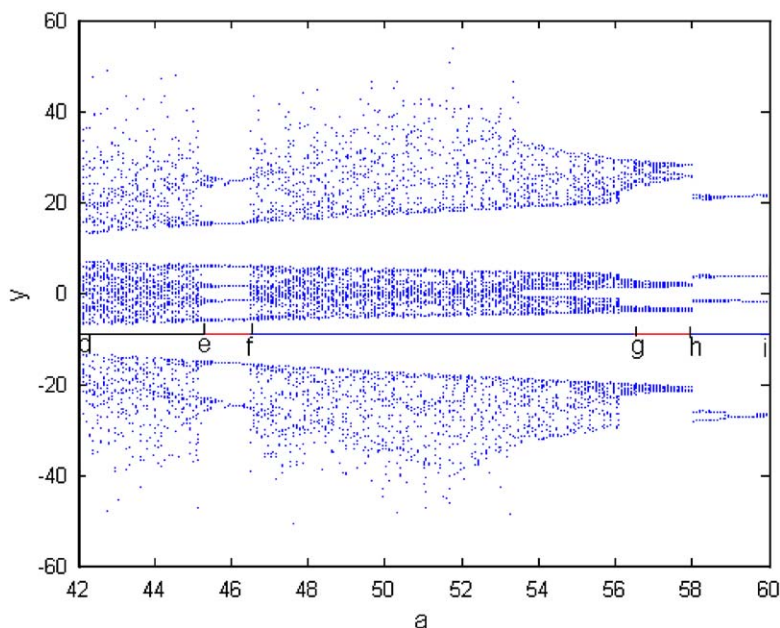


Fig. 12. Bifurcation diagram of y , $b = 8/3$, $c = 104$, $a \in [42, 60]$.

displayed by bifurcation and is identical to the division diagram. Next, the system evolution process is described in more detail. Starting with the first chaotic region d-e, it is demonstrated in Fig. 13(a), where $a = 43$. With a increasing, both Figs. 12 and 13 clearly show the chaos degeneration process in line d-e-f. The system orbit continuously degenerate to the periodic solution region (Fig. 13(c)) through reverse chaos, reverse period-doubling bifurcations (Fig. 13(b), with $a = 45.5$) and periodic solution (Fig. 13, with $a = 46$) in succession, and the periodic orbit abruptly changes to another longer chaotic mode region, as is demonstrated in line f-g of Figs. 11 and 12. Fig. 13(d) shows the phase portrait of $x - z$ with $a = 52$. Then the system gradually goes into periodic orbit again (Fig. 13(f)) through reverse chaos and reverse period-doubling bifurcations (Fig. 13(e)). Notably, to show the rich dynamics of the evolution process, we give different projections of phase portrait or 3D view in Fig. 13.

4. Conclusions

This paper has reported and analyzed a new three-dimensional continuous autonomous chaotic system, in which each equation has a single cross-product term. Basic properties of the system have been analyzed by means of Lyapunov exponents and bifurcation diagrams. Compared with the Lorenz system, it has some distinct differences such as having five equilibria, some larger chaotic regions, and more

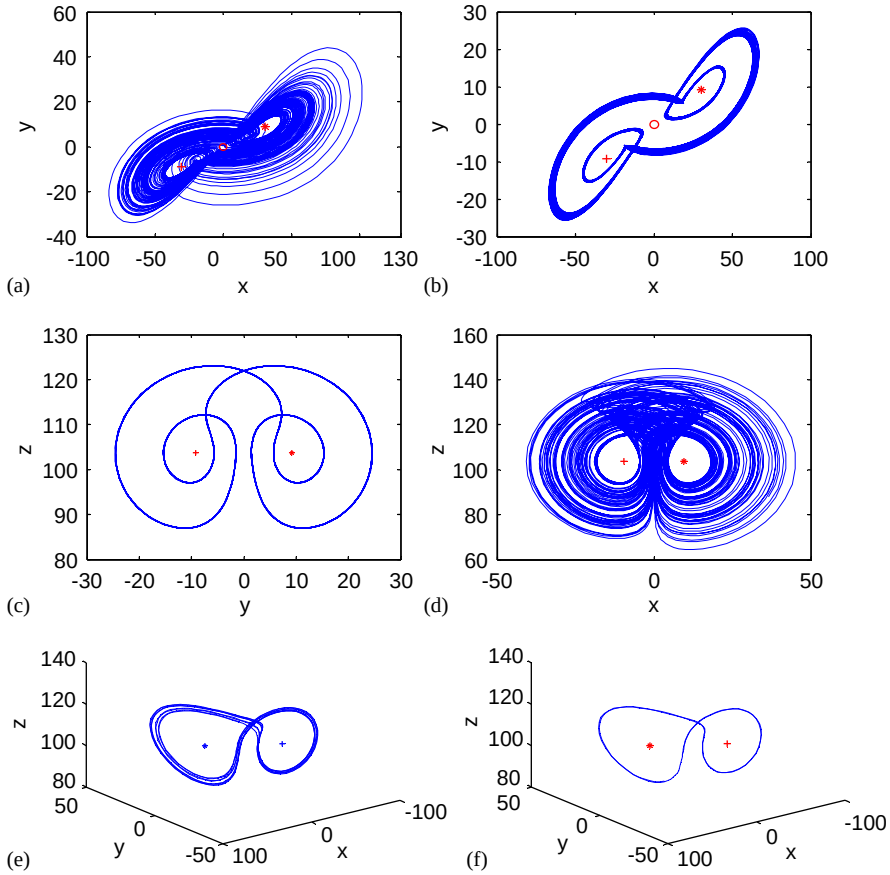


Fig. 13. Dynamical property of evolving process for system (6), $b = 8/3$, $c = 104$, $a \in [42, 60]$, where symbol 'o', '*', and '+' denote equilibria S_1 , S_2 and S_3 respectively, S_1 is omitted in sector figure (c–f). (a) Chaotic attractor, $a = 43$, (b) period-multi-doubling bifurcations $a = 45.5$, (c) $a = 46$, (d) $a = 52$, (e) 3D view on the $x - y - z$ space, $a = 57.5$ and (f) 3D view on the $x - y - z$ space, $a = 59$.

complex bifurcation behaviors. Apparently there are more interesting problems about this new system in terms of dynamics, complexity, control and synchronization, among others, leaving rooms for further studies.

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